TCS CORPORATE WEBSITE

Embed Block — Developer Guide

Role / audience	Development — Developers, technical leads, block owners, AI coding agents
Version	1.0
Status	Draft for review
Owner	Technical Lead / Block Owner (proposed)
Last validated	2026-11-08
Documents	1
Classification	INTERNAL — classification to be confirmed

Generated from the Markdown sources of docs/handover (branch migrated-page-content-edam); do not edit this PDF.

Contents

Embed Block — Developer Guide
- Summary
- Overview
- Source files
- Components, models and filters
- Fields
- Dependencies
- For Developers
- Extending the Block
- Responsive Behavior
- Accessibility
- Data Layer and Analytics
- Performance Considerations
- Security and Privacy
- For Architects
- Troubleshooting
- Known limitations of the block
- Change history of the block
- Reference pages
- Coverage of the required topics
- Screenshots
- Known limitations
- Open items
- Change history

Embed Block — Developer Guide

Attribute	Value
Document ID	DEV-B-embed
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	How the Embed block is built: files, model, structure, behaviour, variations, tests and limitations
Scope	blocks/embed/ and its components embed
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	2026-11-08 (model, files, imports from the code; reference pages from the live site)
Source files	blocks/embed/README.md, blocks/embed/_embed.json, blocks/embed/embed.css, blocks/embed/embed.js
Related documents	Variations, Test coverage, Authoring guide
Related AEM paths	/content/tcs-corp/design-block-master/en/block-library/embed, /content/tcs-corp/design-block-master/en/block-library/embed/embed

1. Summary

Attribute	Value
Block	Embed
Folder	`blocks/embed`
Type	Independent
Category	media
Status	Active
Purpose	Third-party embeds (YouTube, Vimeo, X and other URLs) behind a privacy-friendly facade
Variations (summary)	Provider by URL
Children	—
Code size	179 lines of JS and CSS

2. Overview

Business purpose. Include video and social content hosted on external platforms at a low performance cost (lazy or click-to-load) and with accessible names.

User experience. With a placeholder image visitors see the image with a round Play button; the provider's player loads and starts after Play. Without one, the player or post appears when the block scrolls into view.

Templates / page types. Any page type.

Data source. The block's URI and optional placeholder image; the content itself comes from YouTube, Vimeo, X or the embedded site.

Related blocks. Video (AEM Assets files, captions, transcript, autoplay, background, pop-up and display options); Hero and Teaser (Lottie animations, not embeds).

When to use. A YouTube, Vimeo or X item, or an approved page that allows framing, needing no playback options.

When not to use. AEM Assets videos, captions files, autoplay or background videos, a pop-up player or sizes and aspect ratios (use Video); Instagram, LinkedIn, Facebook or TikTok posts (no dedicated handler; providers usually refuse framing); content whose interactions must be tracked (iframe interactions aren't visible to the page).

Realistic use cases.

- A keynote on YouTube behind a branded placeholder image on a campaign page.
- An official TCS post on X in a press release.
- An approved partner's interactive page in an iframe.

3. Source files

File	Link
`README.md`	GitHub
`_embed.json`	GitHub
`embed.css`	GitHub
`embed.js`	GitHub

4. Components, models and filters

Component	Title	Resource type	Template name	Model	Children allowed
`embed`	Embed	`core/franklin/components/block/v1/block`	Embed	`embed`	—

5. Fields

The Additional tab fields (anchor, hide, padding, margin, animation) are common to the blocks and described in the content modeling guide.

5.1 Embed (`embed`)

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Content	Placeholder Image	`embed_placeholder`	reference	No	—	—	—
Content	Placeholder Image Alt Text	`embed_placeholderAlt`	text	No	`""`	—	—
Content	URI	`embed_uri`	text	No	—	—	—

"Required in model" is the model's `required` flag; whether a field is required in practice is stated in the authoring guide where it is known.

6. Dependencies

Import
`../../scripts/aem.js`
`../../decorators/icon-button-decorator.js`
`../../decorators/media-decorator.js`
`../../utils/media.js`
`../../utils/viewport.js`
`../../utils/placeholders.js`

7. For Developers

```
blocks/embed/
├── README.md
├── _embed.json
├── embed.css
└── embed.js
```

Decorator/Utility	Purpose	Execution or dependency point
scripts/aem.js	Edge Delivery core library (block loading, CSS, metadata, optimized pictures)	`embed.js`: { loadScript }
decorators/icon-button-decorator.js	`createIconButton`	`embed.js`: { createIconButton }
decorators/media-decorator.js	`createFrame`, `createYoutubeFrame`, `createVimeoFrame`, `createVideoElement`	`embed.js`: { createFrame, createVimeoFrame, createYoutubeFrame, }
utils/media.js	`isYoutube`, `isVimeo`, `isAnimation`, `youtubeEmbedUrl`, `vimeoEmbedUrl`, `videoType`, `deliveryAsset` (reads a Dynamic Media with OpenAPI delivery URL of a separate AEM Assets environment), `deliveryImageUrl` (its URL	`embed.js`: { isVimeo, isYoutube }
utils/viewport.js	`onVisible` (once), `watchVisibility` (on / off screen), `currentBreakpoint` (mobile / tablet / desktop)	`embed.js`: { onVisible }
utils/placeholders.js	`getPlaceholders`, `loadPlaceholders`, `fillPlaceholder` (UI text from the locale's placeholders, with English fallbacks)	`embed.js`: { getPlaceholders }

JavaScript (`embed.js`). `decorate` takes the first `picture` (placeholder) and the first link's `href` (else the first `http(s)://` URL in the block's text), empties the block and returns if there is no URL. With a placeholder it builds `div.embed-placeholder` with the picture and a primary icon button (`createIconButton`, `data-cmp-interaction="play"`) calling `loadEmbed(block, link, { autoplay: true, focus: true, t })`; without, `onVisible(block, …)` calls `loadEmbed` without autoplay. `loadEmbed` runs once (`embed-is-loaded` guard), picks the first matching entry of `EMBEDS` (`youtube`, `vimeo`, `twitter`) or `createFrame`, replaces the block's children and adds `embed-{name}` and `embed-is-loaded`. The module exports only `decorate`.

CSS (`embed.css`). `max-width: 56.25rem`, centred; the block reserves 16:9 while it has neither a placeholder nor a loaded embed; `.embed-frame` and `.embed-placeholder` are 16:9 with `--radius-l`.

Input DOM. Not key-value: AEM delivers an empty row per dialog tab (Content, Additional) and one cell with the grouped fields (picture with the collapsed alt text, then the URI as a link or text).

Rendered DOM contract. Before loading: `.embed > .embed-placeholder > picture + button.embed-placeholder-play`, or an empty `.embed`. After: `.embed.embed-is-loaded[.embed-youtube| .embed-vimeo|.embed-twitter] > .embed-frame > iframe` or `> blockquote.twitter-tweet > a`.

Events and async behavior. `getPlaceholders` is awaited after the URL check; IntersectionObserver (`onVisible`); `loadScript('https://platform.twitter.com/widgets.js')` for X posts.

Error handling. No URL: empty block. Provider failures are shown by the provider; the block logs nothing.

Parent / child contracts. Tabs (`blocks/tabs/tabs.js`) posts pause commands to YouTube (`enablejsapi=1`) and Vimeo frames of hidden tabs.

8. Extending the Block

1. **A new provider:** add an entry `{ name, match, embed }` to `EMBEDS` in `embed.js`, building the DOM with the media decorator (`createFrame`) or DOM APIs, never HTML strings; add styles scoped as `.embed.embed-{name}`.
2. **A dialog field:** add it to `_embed.json` (it's a grouped `embed_*` cell: read it by element, not by position), then `npm run build:json` and `npm run lint`. Field names ending in Text, Title, Type, Alt or MimeType collapse into their base field.
3. **Responsive / accessibility:** keep frames titled from placeholders, focus after Play and the reserved aspect ratio.
4. **Data layer:** describe new controls with `data-cmp-interaction`.
5. **Tests:** URL parsing belongs in `utils/media.js` with tests in `test/media/`.
6. **Backward compatibility:** the block reads links and pictures anywhere in the block, so tab rows and the order of the grouped fields don't matter; keep it that way.
7. **Library and docs:** update the block library, this README, `npm run build:docs`; `npm run build:modules` after changing imports.

9. Responsive Behavior

`embed.css` has no media queries.

Viewport	Layout	Image behavior	Interaction behavior
All viewports	Up to 56.25rem wide, centred; 16:9 frames and placeholder with `--radius-l` corners; X posts at the widget's size, centred	Placeholder image covers the 16:9 area (`object-fit: cover`)	Play (scaled 1.4×) centred on the image; same behaviour everywhere

10. Accessibility

- **Semantics:** iframes with a `title`; a native `button` for Play; X posts as a `blockquote` with a link.
- **Headings:** none in the block; add one before it.
- **Keyboard and focus:** Play is reachable with Tab and activated with Enter / Space; after Play focus moves into the iframe (or the X link).
- **Accessible names:** Play from the placeholder alt text; frames "Content from {host}".
- **Moving content:** videos start only after Play (click-to-load) or not at all; a video in a hidden tab of a Tabs Container is paused (YouTube / Vimeo pause commands).
- **Reduced motion:** the Additional tab animation doesn't play; embeds never autoplay without Play.
- **Contrast and reflow:** the block is fluid; the Play button uses the primary icon button styles.
- **Empty / loading / error states:** empty without a URL; the reserved area while loading; provider error pages inside the frame.

Author responsibilities: placeholder alt text, a heading before the block, videos with captions on their platform, a transcript in the page for informative videos.

Developer responsibilities: keep the frame names and the Play label from placeholders, the focus move and the DOM-built players.

Key	Expected behavior
Tab	Moves to the Play button, then into the player once loaded
Enter / Space	Activates Play: the embed loads and receives focus

11. Data Layer and Analytics

`decorators/data-layer-decorator.js` adds the block as `tcs-corp/components/embed`, id = Anchor ID else `embed-{n}`. The block sets `data-cmp-interaction="play"` on the Play button.

Trigger	Data-layer event	Required properties	Analytics behavior
Block decorated	component state	`@type`, `parentId`	One entry per block
Play button click	`cmp:click`	`interaction` `play`, `dc:title` (the button's name), `breakpoint`	A play request
X fallback link click	`cmp:click`	`interaction` `link`, `xdm:linkURL`, `xdm:linkType` `exit`	Exit link

Interactions inside the provider frames and the X widget aren't visible to the page. The intended path is Adobe Client Data Layer, then Adobe Data Collection (Tags), the AEP Web SDK, the Edge Network and Analytics / CJA, but Tags and the Web SDK aren't configured yet (blocked: measurement plan); no consent mechanism is implemented.

12. Performance Considerations

- Decorated with its section; only placeholders are fetched before the URL is used.
- No provider request until the block is in view (no placeholder) or until Play; iframes `loading="lazy"`; `widgets.js` only when an X post loads.
- 16:9 space reserved before loading (no layout shift for videos and iframes); X posts change height when the widget renders.
- One IntersectionObserver per block, disconnected after loading.
- A placeholder image in the first section may be the LCP element.

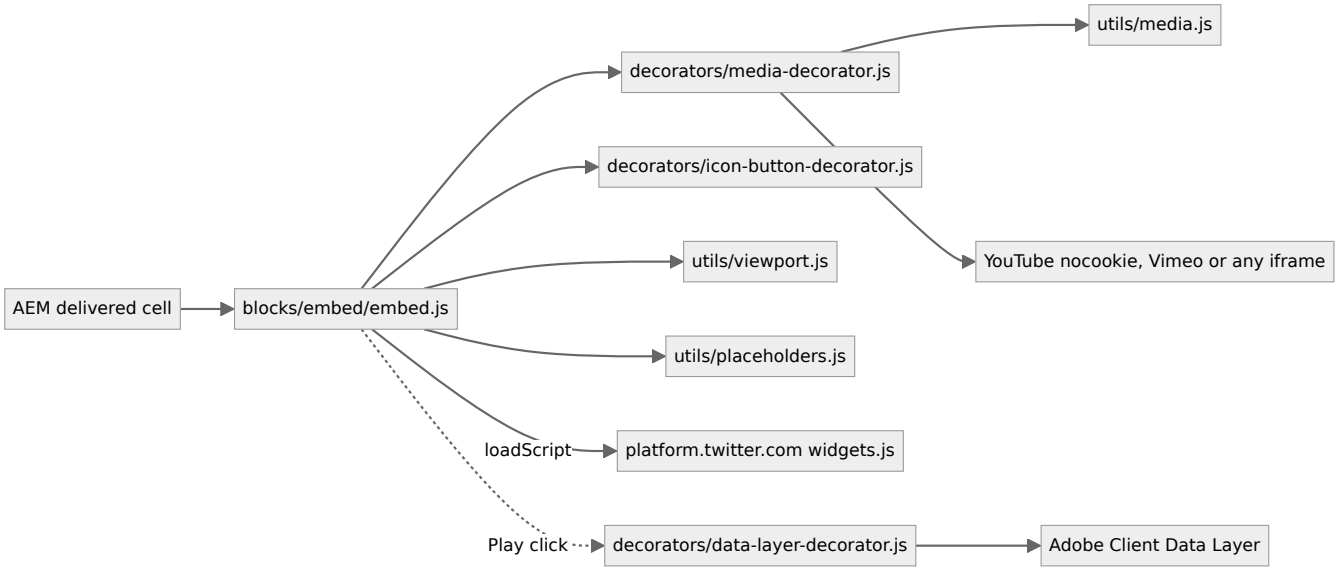
13. Security and Privacy

- Players are built with DOM APIs (`decorators/media-decorator.js`); no authored URL is written as HTML.
- Any URL becomes a generic iframe: there is no host allow-list and no `sandbox` attribute; only approved, framable URLs should be authored.
- YouTube always uses `www.youtube-nocookie.com`, Vimeo `dnt=1`; the X widget and generic iframes load third-party content and cookies as their providers do. No consent mechanism is implemented.
- `x.com` is rewritten to `twitter.com` for the widget; other query strings of YouTube URLs aren't passed on (only the id is used).

14. For Architects

- **Role:** a thin client-side wrapper for provider embeds.
- **Content model:** one grouped cell (placeholder, alt, URI) plus the Additional tab.
- **Rendering:** DOM-built iframes via `decorators/media-decorator.js`; X via the provider widget.
- **Dependencies:** media decorator / utils, icon button decorator, viewport and placeholders utils, `loadScript` from `scripts/aem.js`.
- **Data layer / Web SDK / Target:** generic component state and the Play click; no Web SDK or Target.
- **Consent and identity:** none implemented; YouTube privacy-enhanced host and Vimeo `dnt=1` are always used; the X widget and generic iframes set whatever the provider sets once loaded.
- **Caching / performance:** page HTML and image via the CDN; provider content cached by providers; nothing loads before visibility or Play.

- **Security:** see below.



15. Troubleshooting

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Nothing renders	No URI	`{page}.plain.html`: look for the link or URL	Paste the full https URL	Embed appears
YouTube "Video unavailable"	Wrong id, private video, embedding disabled	Inspect the iframe `src`	Public, embeddable video URL	Player loads
Vimeo "does not exist"	URL without a numeric id	Inspect the iframe `src`	Use `vimeo.com/{id}`	Player loads
X post shows only a link	Widget script blocked, post protected or deleted	Network: `widgets.js`; open the post	Public post; allow the script	Card renders
Iframe "refused to connect"	Site forbids framing, or http	Console (X-Frame-Options / CSP)	Embeddable https URL, or a link instead	Page shows
Video starts with sound after Play	`muted=1` is passed to YouTube as well; browsers' autoplay rules apply	Inspect the `src` parameters	Expected; use Video for playback control	—
Placeholder cropped badly	Image not 16:9	Compare dimensions	16:9 rendition, subject centred	Image framed well

Evidence to collect (no secrets): page URL, `.plain.html` of the block, the iframe `src`, console errors (CSP, framing), Network entries of the provider requests, the block's data-layer entries.

16. Known limitations of the block

Limitation	Impact	Workaround	Planned correction
No start time, playlist or other YouTube parameters	Plays from the start	Use the Video block (Start At)	Not recorded
Generic iframes always 16:9	Tall pages are cut off	Link to the page instead	Not recorded
No host allow-list for generic iframes	Any framable URL can be embedded	Governance of who authors Embeds	Not recorded

Limitation	Impact	Workaround	Planned correction
No handlers for Instagram, LinkedIn, Facebook, TikTok	Usually refused as iframes	Link to the post	Not recorded
Iframe interactions not tracked	No provider play data	None	Not recorded
Frame name is generic ("Content from {host}")	Screen readers don't hear the video's name from the frame	Heading before the block; placeholder alt text	Not recorded
`muted=1` passed to YouTube after Play	The video may start with sound	Use the Video block	Not recorded
An Embed without a URL still reserves an empty 16:9 area	Blank space on the page	Delete the block or add a URI	Not recorded
No consent management	Provider cookies once loaded	Click-to-load placeholder	Not recorded (owner: privacy / legal)

17. Change history of the block

Version	Change	Source	Date
Not recorded	README generated from verified implementation and approved handover documentation	Repository review	Not recorded
Handover 2.18	Additional tab content animation (Animation Type, Motion Speed, Animation Trigger)	docs/handover revision history	18 October 2026
Handover 2.27	Block audit: Play named after the placeholder alt text (D-D-8), bare URLs rendered (D-D-9), privacy-enhanced players (D-D-10)	docs/handover revision history; docs/audit/BLOCK-ECOSYSTEM-AUDIT.md	19 October 2026
Handover 2.29	Re-audit: Vimeo player / channel / unlisted URLs (R-D-5); embeds paused in hidden tabs (R-C-8)	docs/handover revision history; docs/audit/REAUDIT-2026-10-20.md	20 October 2026

18. Reference pages

Kind	Page	Live URL
Block library	/design-block-master/en/block-library/embed	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/embed
Block library	/design-block-master/en/block-library/embed/embed	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/embed/embed

19. Coverage of the required topics

Topic	Where
Purpose	Overview
Source files	generated: Files
Model, definition, filters, children	generated: Components
Dependencies	generated: Dependencies
Authored input / rendered output / DOM transformation	For Developers
Configuration fields, defaults, required fields	generated: Fields
Variations	EMBED-VARIATIONS.md
Responsive behaviour	Responsive Behavior

Topic	Where
Accessibility and keyboard	Accessibility
Analytics / data layer	Data Layer and Analytics
Performance	Performance Considerations
Security	Security and Privacy
Extending	Extending the Block
Known limitations	Known Limitations
Troubleshooting	Troubleshooting
Tests	EMBED-TEST-COVERAGE.md
SEO behaviour	Not recorded
Caching considerations	Not recorded (Edge Delivery default caching; no block-specific rule)
Ownership	Technical Lead / Block Owner (proposed)

20. Screenshots

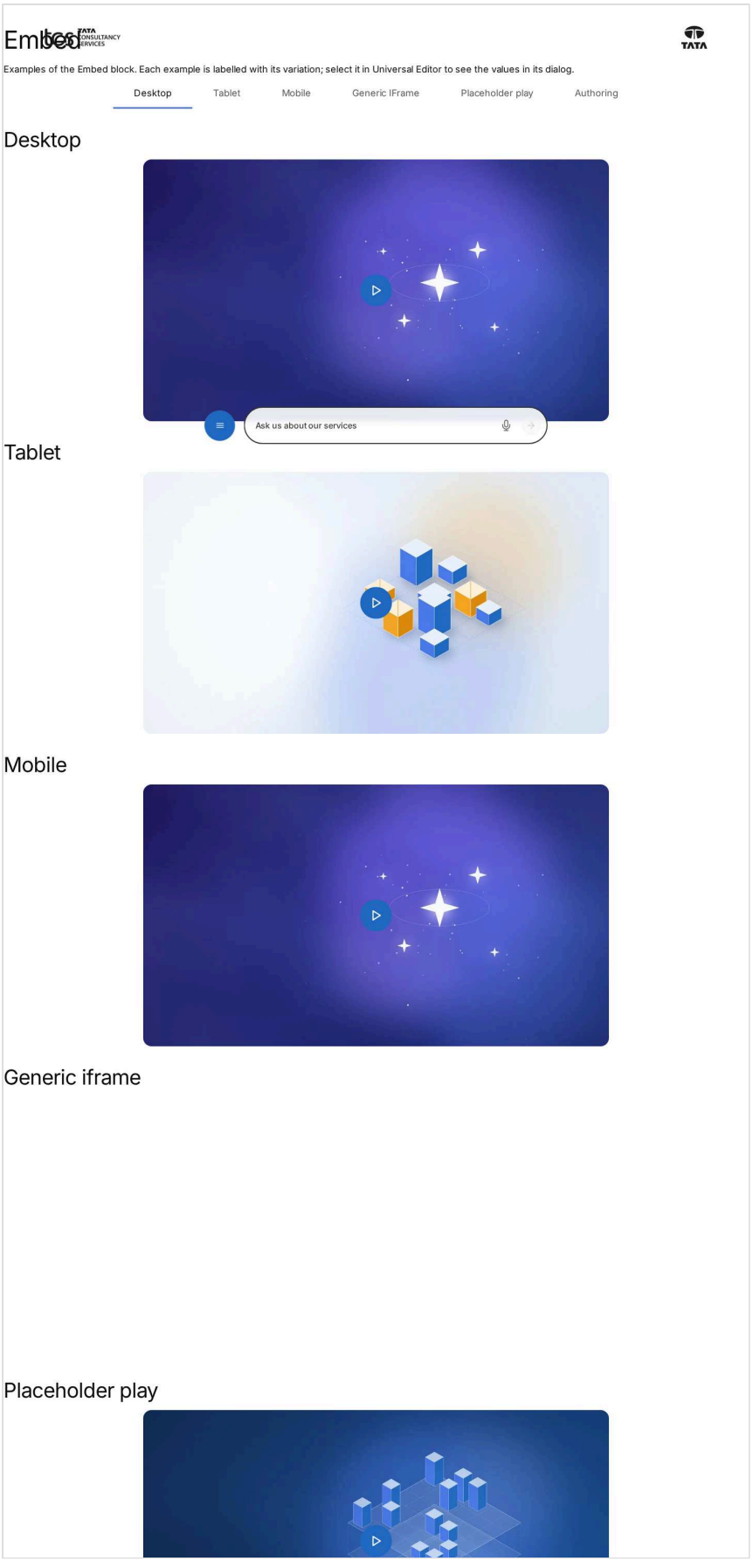


Figure: Embed block library page — desktop 1440 × 900 (/design-block-master/en/block-library/embed/embed, captured 2026-11-08).

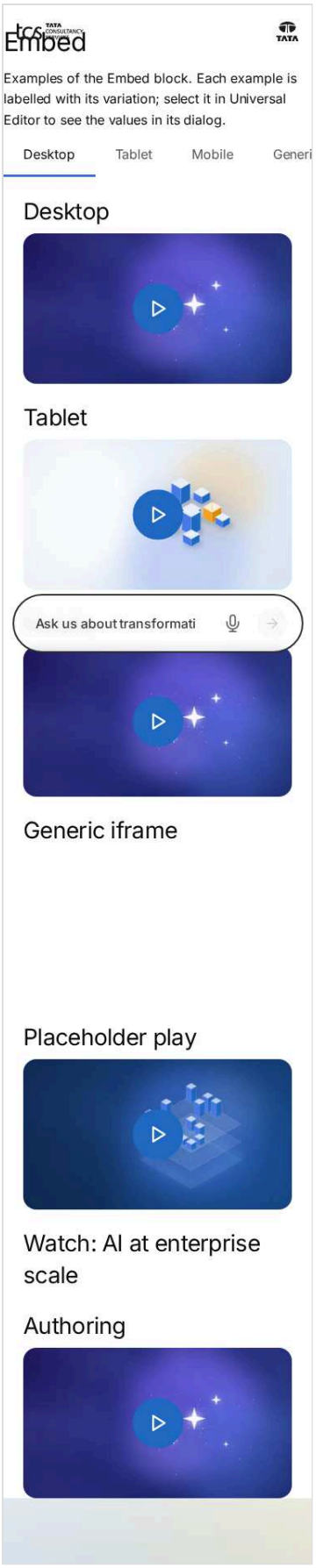


Figure: Embed block library page — mobile 375 × 812 (/design-block-master/en/block-library/embed/embed, captured 2026-11-08).

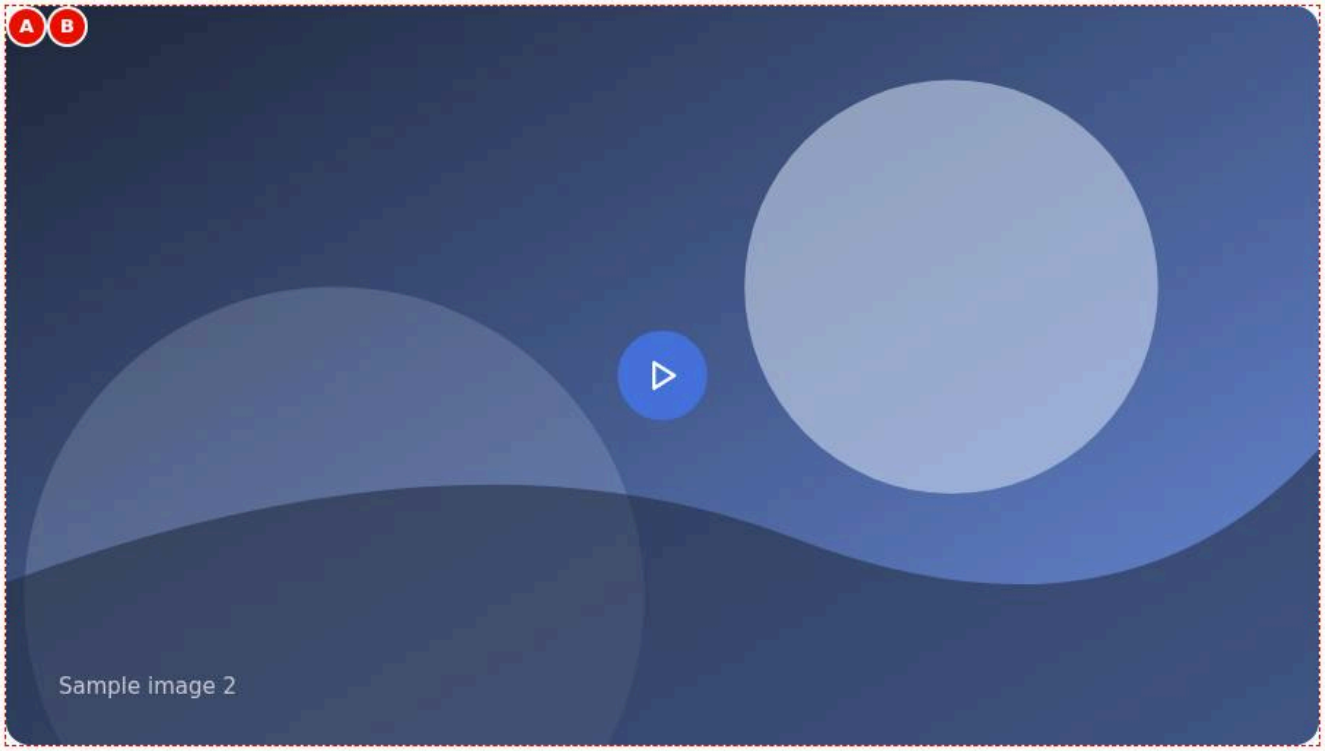


Figure: Desktop rendering — The Embed at 1440 px: the placeholder image photo-2.svg in a 16:9 frame 900 px wide (56.25rem), centred, with 1rem rounded corners, and the round primary Play button in its centre. The YouTube player is not loaded yet. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/embed-desktop-view.jpg) (handover screenshot; rendering at the time it was taken).

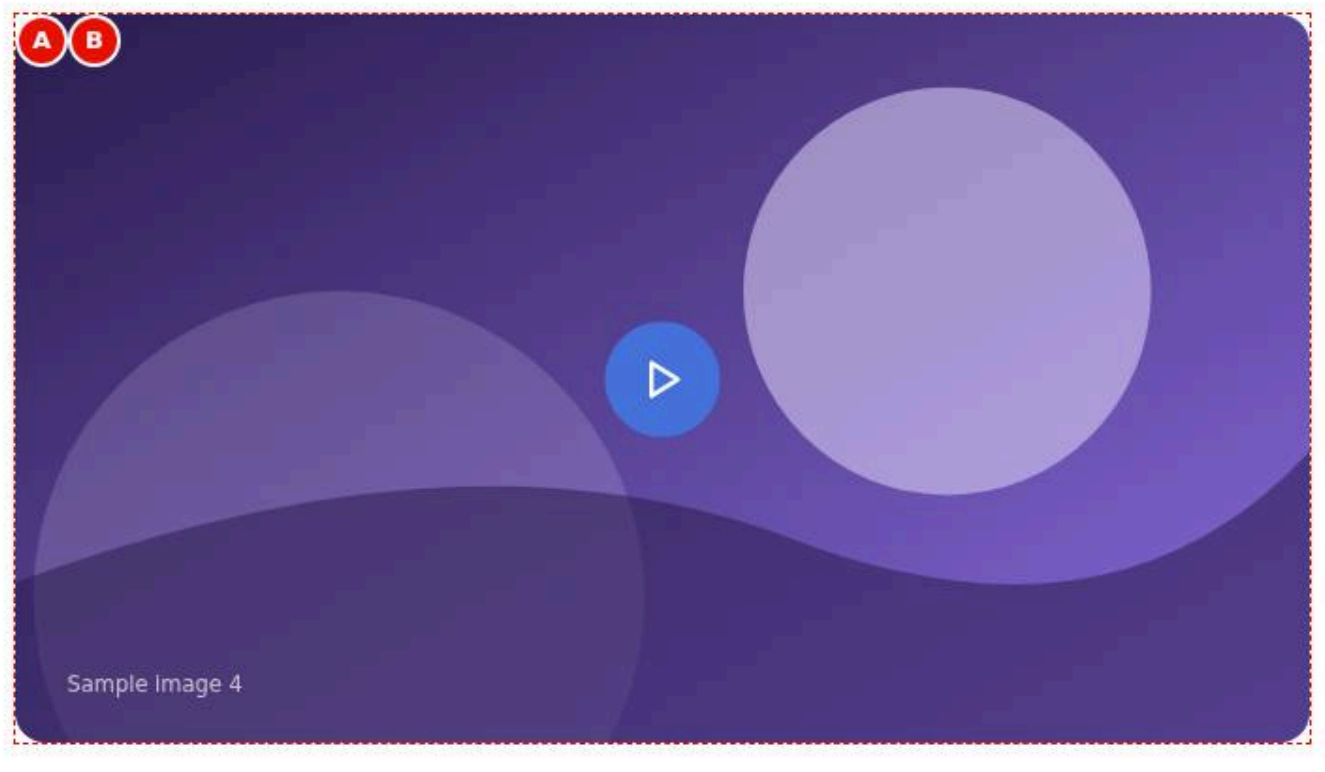


Figure: Tablet rendering — The Embed at 768 px: the placeholder image photo-4.svg at the full content width (about 688 px) in a 16:9 frame with rounded corners and the Play button centred; the Vimeo player

loads only after Play. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/embed-tablet-view.jpg) (handover screenshot; rendering at the time it was taken).

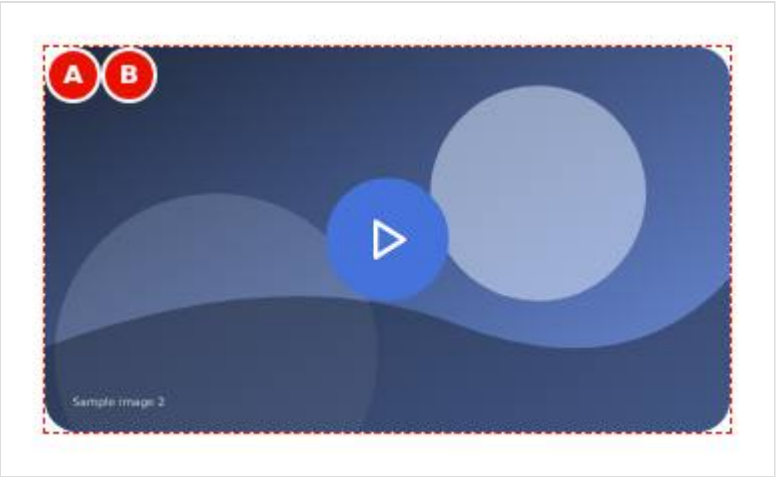


Figure: Mobile rendering — The Embed at 390 px: the placeholder image photo-2.svg at the full content width (about 342 px) in a 16:9 frame with rounded corners and the Play button centred. No X post is part of this example. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/embed-mobile-view.jpg) (handover screenshot; rendering at the time it was taken).



Figure: Variation example — The Embed at 1440 px: a 16:9 iframe 900 px wide with rounded corners showing a partner's Market explorer page with Region and Industry filters and the figures +6.4% digital spend growth, 38% cloud adoption and 2.1x AI investment since 2022. The capture uses a local stand-in page for the partner URL. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/embed-generic-iframe.jpg) (handover screenshot; rendering at the time it was taken).

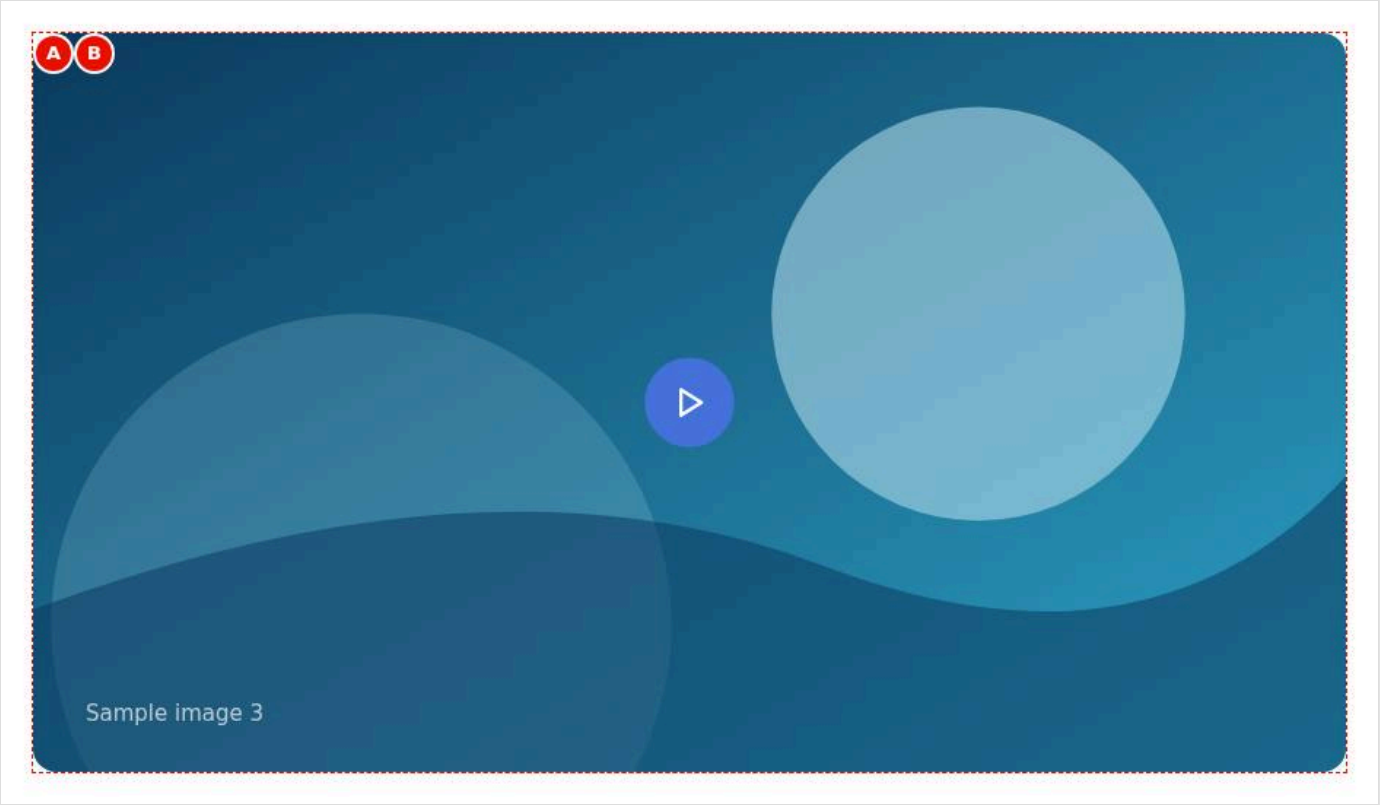


Figure: Variation example — The Embed at 1440 px: the placeholder image photo-3.svg in a centred 900 px 16:9 frame with rounded corners and the round primary Play button in the centre. Only the state before Play is shown; the YouTube player is not in the screenshot. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/embed-placeholder-play.jpg) (handover screenshot; rendering at the time it was taken).

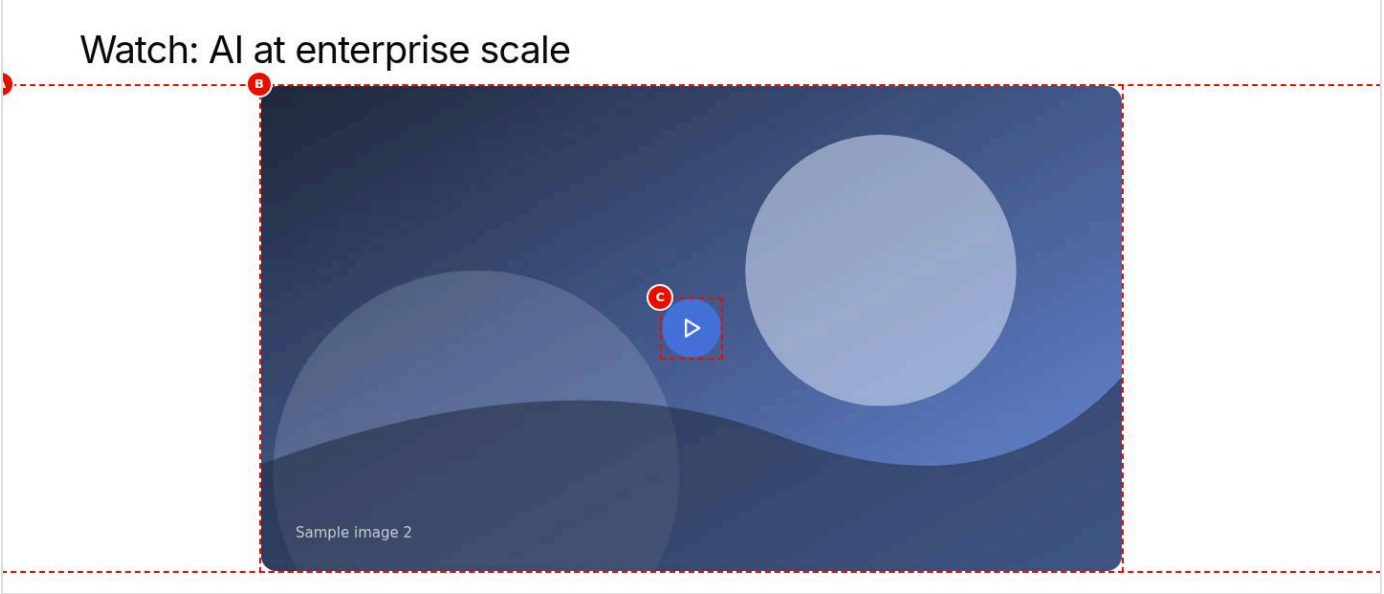


Figure: Universal Editor canvas — An annotated edit-mode view at 1440 px (not a Universal Editor screenshot): the H2 Watch: AI at enterprise scale above the Embed block (A) with the placeholder image photo-2.svg (B) and the generated Play button (C) centred on it. No properties panel is shown. (Annotated preview in editor mode with sample content; letters as in the annotations list, not a Universal Editor screenshot; file screenshots/embed-authoring-view.jpg) (handover screenshot; rendering at the time it was taken).

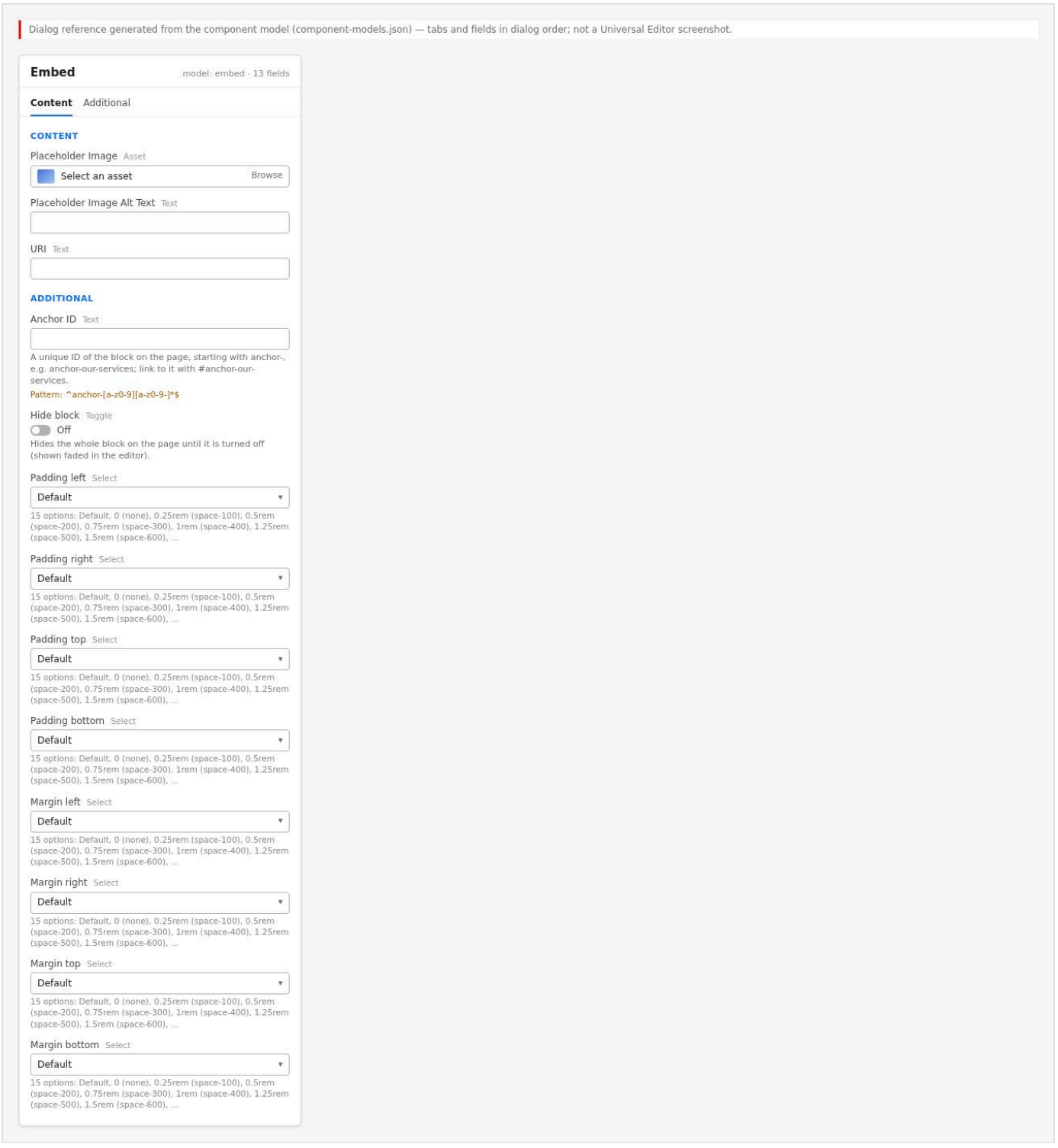


Figure: Properties panel — The Embed properties panel: Content tab with Placeholder Image, Placeholder Image Alt Text and URI; the Additional tab. (Dialog reference generated from the component model, not a Universal Editor screenshot; file screenshots/embed-properties-panel.jpg) (handover screenshot; rendering at the time it was taken).

21. Known limitations

- Universal Editor dialogs were not captured: no author access in this program (see the access blockers).

22. Open items

- None.

23. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages